Biosynthesis of Cholesterol

Cholesterol

- The level of cholesterol in blood is related to the development of atherosclerosis & MI.
- Cholesterol is found exclusively in animals, hence it is often called as animal sterol.
- The total body content of cholesterol in an adult man weighing 70 kg is about 140 g

- All steroids have cyclopentanoperhydrophenanthrene ring system.
- Cholesterol is amphipathic in nature.
- It possesses both hydrophilic & hydrophobic regions in the structure.
- All carbon atoms of cholesterol are derived from acetyl CoA.

- Total 27 carbon atoms.
- One hydroxyl group at 3rd position which is characteristic of all sterols.
- The OH group is β-oriented, projecting above the plane of ring.
- Double bond between carbon atoms 5 & 6.
- An eight carbon side chain, β-oriented, attached to 17th carbon.

Sites

- Major sites are liver, adrenal cortex, testes, ovaries & intestine.
- All nucleated cells can synthesize cholesterol, including arterial wall.
- Location:
- The enzymes involved in the synthesis of cholesterol are partly located in endoplasmic reticulum & partly in cytoplasm.

Requirement

- Acetyl CoA provides all carbon atoms.
- Reducing equivalents are supplied by NADPH.
- ATP provides energy.
- For production of one molecule of cholesterol…
- 18 moles of acetyl CoA
- 36 moles of ATP
- 16 moles of NADPH are required

Stages

- **Five stages.**
- **Synthesis of HMG CoA (β-hydroxy β- methylglutaryl CoA)**
- **Formation of Mevalonate**
- **Production of isoprenoid units**
- **Synthesis of squalene**
- **Conversion of squalene to cholesterol.**

Synthesis of (HMG CoA)

- Two moles of acetyl CoA condense to form acetoacetyl CoA.
- Another molecule of acetyl CoA is then added to produce HMG CoA.
- These reactions are similar to that of ketone body synthesis.

- The two pathways are distinct.
- Ketone bodies are produced in mitochondria while cholesterol synthesis occurs in cytosol.
- There exist two pools of HMG CoA in the cell.

- Two isoenzymes of HMG CoA synthase are known.
- The cytosomal enzyme is involved in cholesterol synthesis whereas the mitochondrial HMC CoA synthase participates in ketone body formation.

Formation of mevalonate

- **HMG CoA reductase is the rate limiting enzyme in cholesterol biosynthesis.**
- **This enzyme is present in endoplasmic reticulum & catalyses the reduction of HMC CoA to mevalonate.**
- **The reducing equivalents are suppplied by NADPH.**

Production of isoprenoid units

- In a three step reaction catalyzed by kinases, mevalonate is converted to 3- phospho 5-pyrophosphomevalonate which on decarboxylation forms isopentenyl pyrophosphate (IPP).
- It is isomerizes to dimethylallylpyrophosphate

Synthesis of squalene

- IPP & DPP condense to produce a 10-carbon geranyl pyrophosphate (GPP).
- Another molecule of IPP condenses with GPP to form a 15-carbon farnesyl pyrophosphate (FPP).
- Two units of farnesyl pyrophosphate unite & get reduced to produce a 30-carbon squalene.

Cyclization

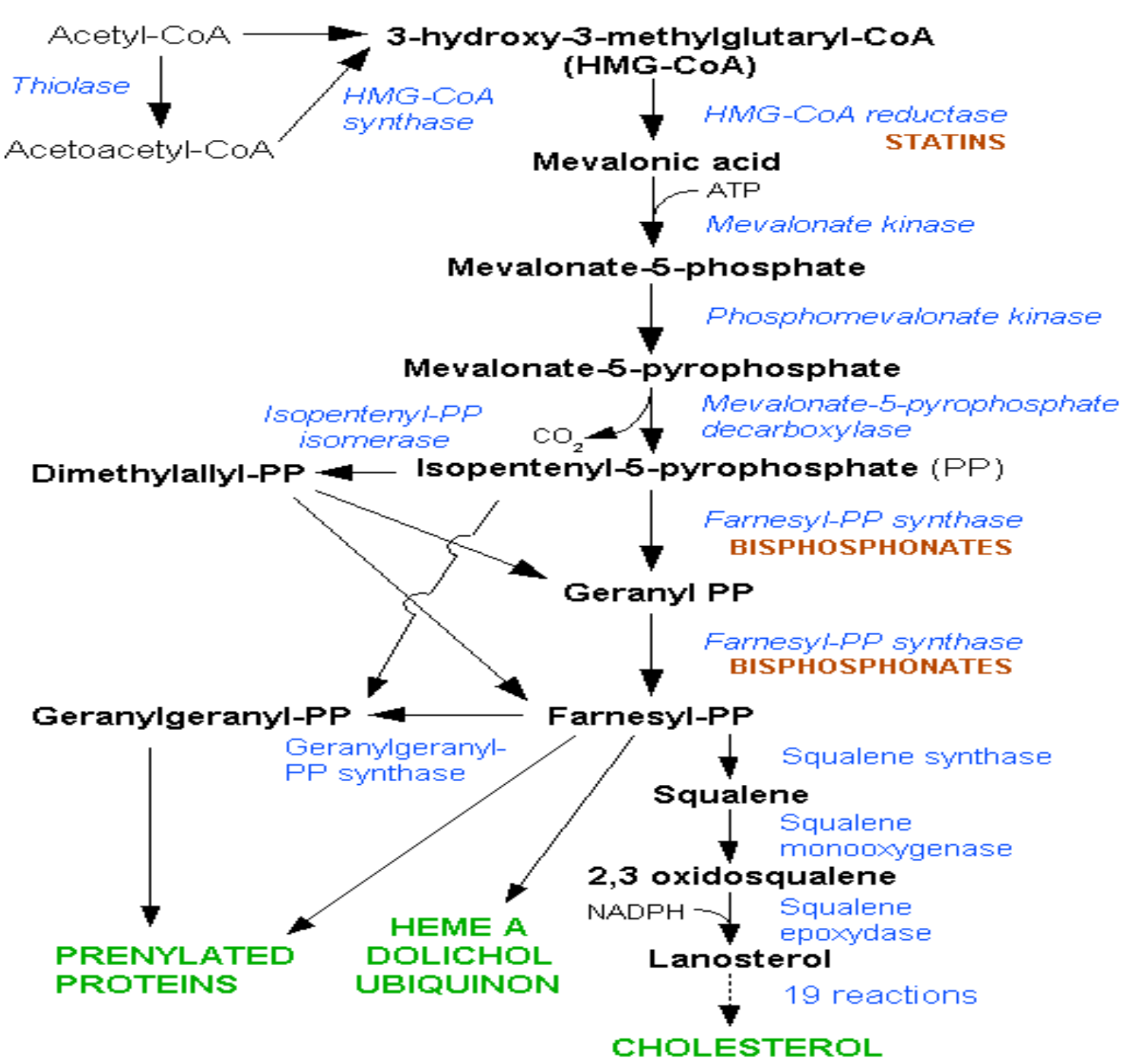
- **Conversion of squalene to cholesterol:**
- **Squalene undergoes oxidation by epoxidase, using molecular oxygen & NADPH to form squalene epoxide.**
- **Cyclase converts it to 30 carbon lanosterol.**
- **It is the first steroid compound synthesized.**

Cutting to size

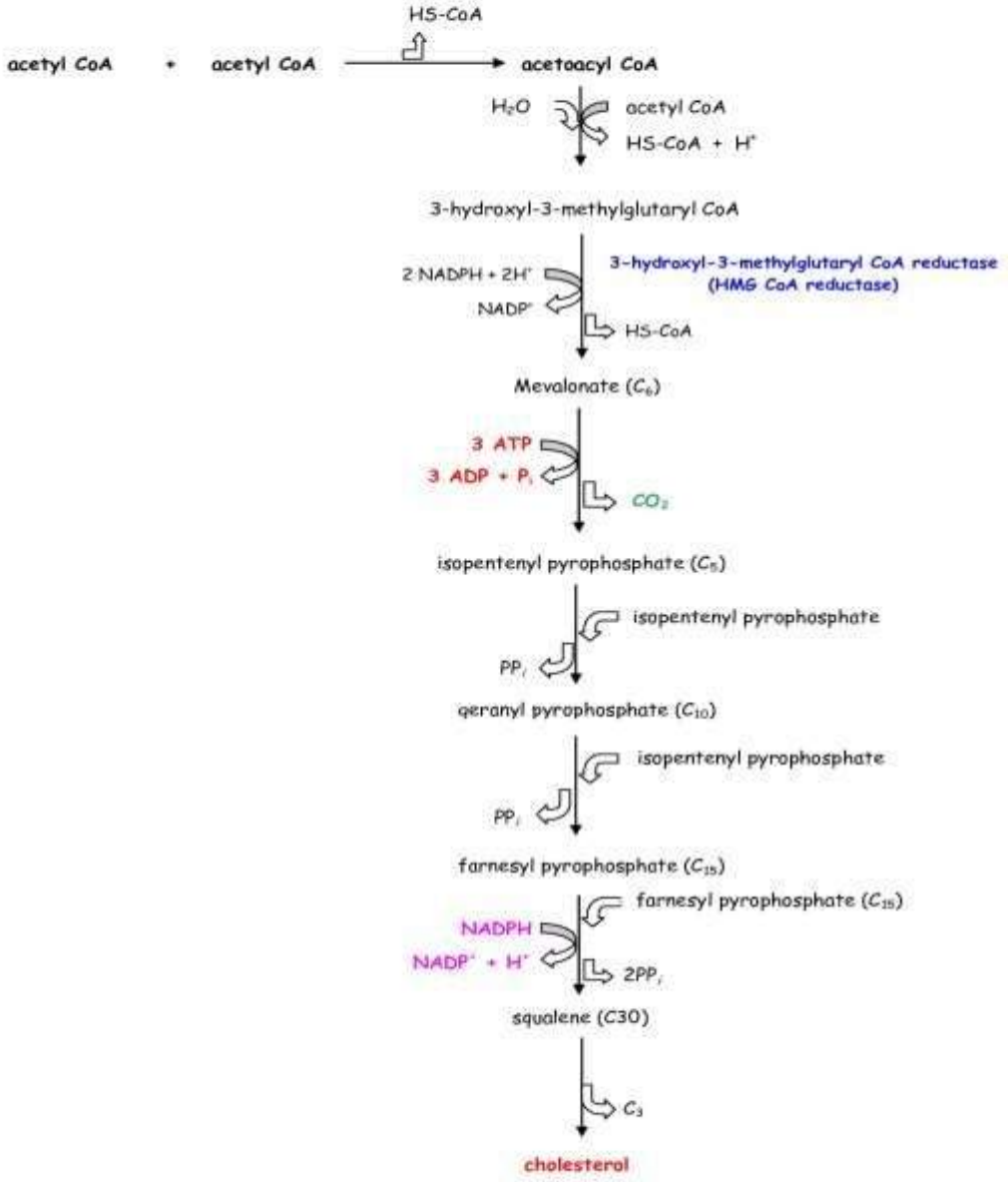
- The formation of cholesterol from lanosterol is a multistep process with a series of about 19 enzymatic reactions.
- Most important reactions:
- Reducing the carbon atoms from 30 to 27.
- Removal of 2 methyl groups from C_4 & 1 methyl group from C_{14} to produce

- Shift of double bond from C_8 to C_5
- Reduction in the double bond present between C_{24} and C_{25}.
- The enzymes (about 19) involved in the Conversion of lanosterol to cholesterol are associated with endoplasmic reticulum.

- **14-desmethyl lanosterol, zymosterol, cholestadienol & desmosterol are among the intermediates in the cholesterol biosynthesis.**
- **The penultimate product is 7-dehydrocholesterol which, on reduction, finally yields cholesterol.**


Acetyl-CoA
3-hydroxy-3-methylglutaryl-CoA (HMG-CoA)
Thiolase
HMG-CoA synthase
Acetoacetyl-CoA
HMG-CoA reductase
STATINS
Mevalonic acid
ATP
Mevalonate kinase
Mevalonate-5-phosphate
Phosphomevalonate kinase
Mevalonate-5-pyrophosphate
Isopentenyl-PP isomerase
CO2
Mevalonate-5-pyrophosphate decarboxylase
Dimethylallyl-PP
Isopentenyl-5-pyrophosphate (PP)
Farnesyl-PP synthase
BISPHOSPHONATES
Geranyl PP
Farnesyl-PP synthase
BISPHOSPHONATES
Geranylgeranyl-PP
Farnesyl-PP
Geranylgeranyl-PP synthase
Squalene synthase
Squalene
Squalene monooxygenase
2,3 oxidosqualene
NADPH
Squalene epoxydase
Lanosterol
19 reactions
CHOLESTEROL
PRENYLATED PROTEINS
HEME A
DOLICHOL
UBIQUINON

Cholesterol Synthesis



Regulation of cholesterol synthesis

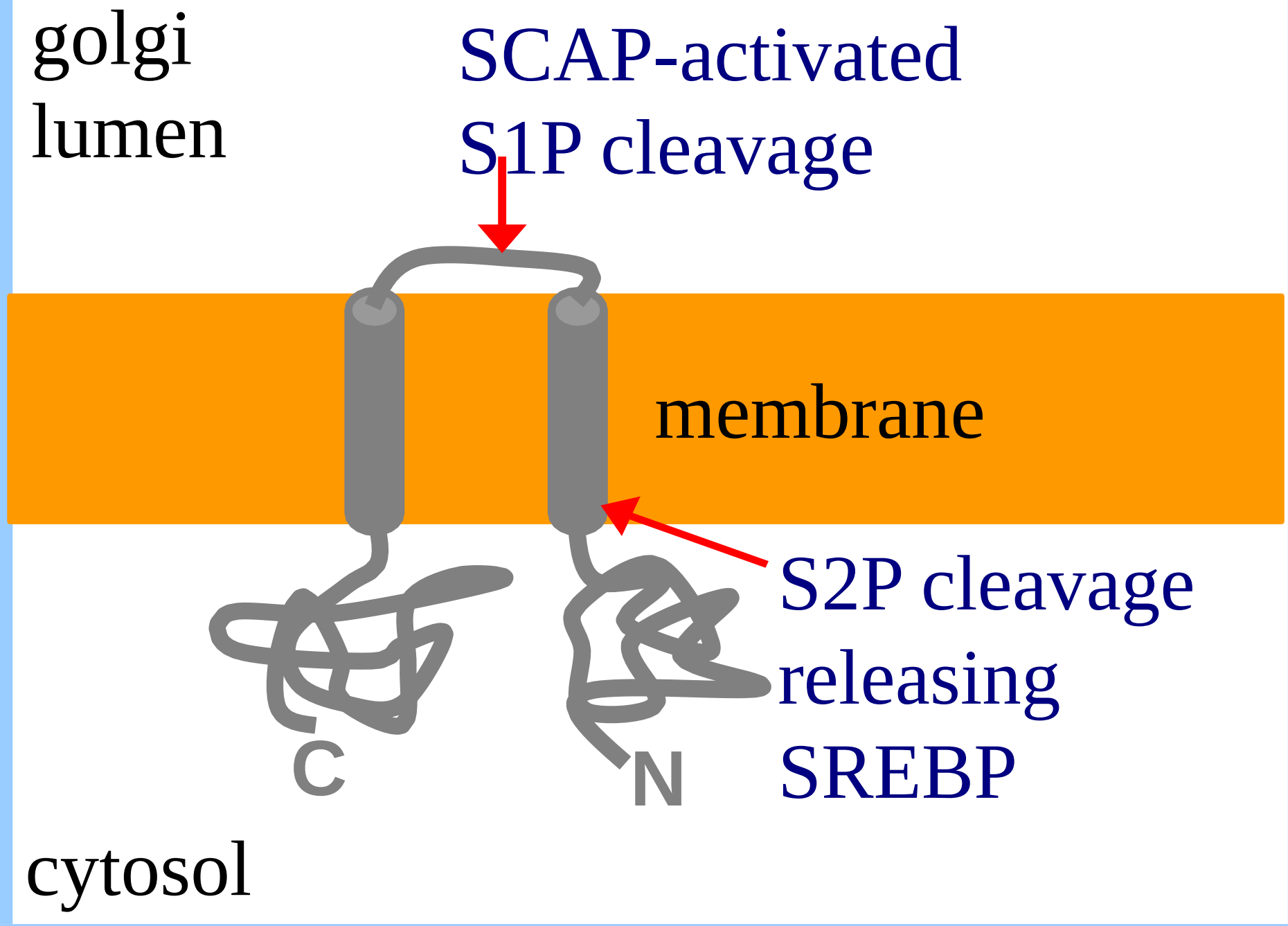
- **HMG CoA reductase is rate-limiting enzyme.**
- **HMG CoA reductase is found in association with endoplasmic reticulum & is subjected to different metabolic controls.**
- **Regulation at transcription:**
- **Long-term regulation involves regulation of transcription of gene for HMG CoA**

- Sufficient cholesterol is present in the cell, transcription of the gene for HMG CoA reductase is suppressed & cellular synthesis of cholesterol is decreased.
- Cholesterol in diet is low, synthesis is increased.
- Cholesterol regulates the expression of HMG CoA reductase gene & LDL receptor gene.

- A specific recognition sequence known as sterol regulatory element (SRE) is present in DNA.
- SRE binding by sterol regulatory element binding protein (SREBP) is essential for the transcription of these genes.
- When cholesterol levels are high, the SREBP remains as inactive precursor.

- SREBP cleavage activator protein (SCAP), is an intracellular cholesterol sensor.
- When cholesterol levels are less, SCAP escorts SREBP to Golgi bodies.
- Two Golgi proteases - site 1 protease 1 & 2 (S1P & S2P) sequentially cleave the SREBP to a protein which binds to SRE & activates transcription of HMG CoA

- **Protease S1P (site 1 protease):**
- **An integral protein of Golgi membranes, cleaves the SREBP precursor at a site in the lumenal domain.**
- **An intramembrane zinc metalloprotease domain of another golgi protease S2P then catalyzes cleavage within the transmembrane segment of the SREBP precursor, releasing SREBP to the cytosol.**
- **Only the product of S1P cleavage can serve as a substrate for S2P.**


golgi
lumen
SCAP-activated
S1P cleavage
membrane
S2P cleavage
releasing
SREBP
C
N
cytosol

Short-term regulation

- **Covalent modification:**
- **HMG CoA reductase is inhibited by phosphorylation, catalyzed by AMP-dependent protein kinase (which also regulates fatty acid synthesis & catabolism).**
- **Dephosphorylation by protein phosphatase 1 makes it active.**

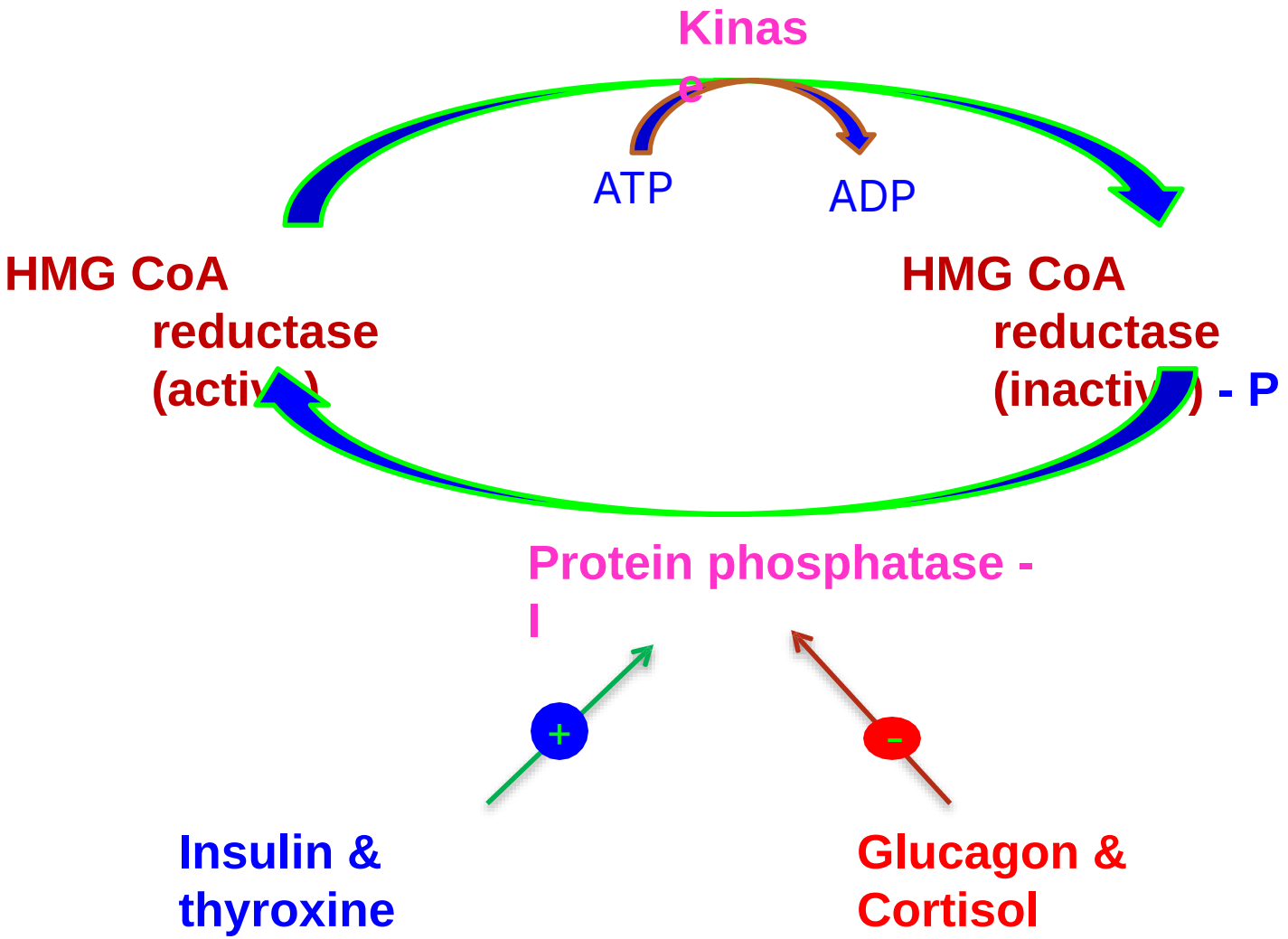
Hormonal regulation

- Insulin & thyroxine increases the activity of HMG CoA reductase.
- Cortisol & glucagon decreases the activity of HMG CoA reductase.
- HMG CoA reductase activity is inhibited by bile acids.
- Fasting is also reduces the activity of HMG CoA reductase.

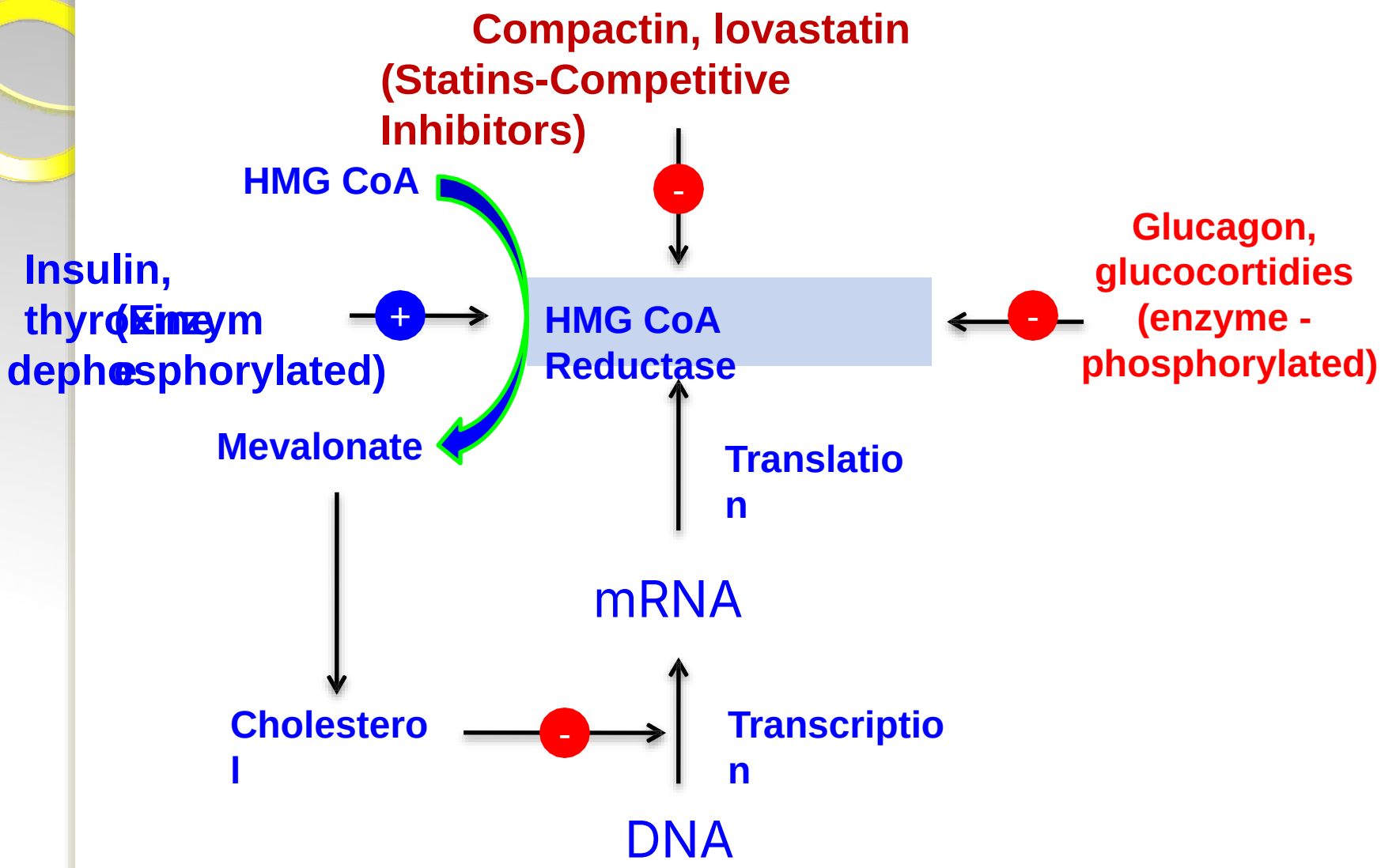
Drugs

- Lovastatin & other statin group drugs are **competitive inhibitors of HMG CoA reductase.**
- These drugs are used to **reduce the cholesterol levels in blood.**

Activation of HMG CoA reductase



Regulation of cholesterol biosynthesis by HMG CoA reductase



Cholesterol pool

- The total body cholesterol content varies below 200 mg/dl.
- LDL transports cholesterol from liver to peripheral tissues.
- HDL transports cholesterol from tissues to liver.
- Cells of extrahepatic tissues take

- **The free cholesterol released within the cell has following fates:**
- **Incorporated into cell membranes.**
- **Metabolized to steroid hormones, especially in adrinal cortex & gonads**
- **Esterified with saturated fatty acids & stored in cells.**

- The enzyme ACAT (acyl cholesterol acyl transferase) helps in this reaction.
- Esterified with LCAT & incorporated into HDL, trnasported & finally excreted through liver.

Excretion of cholesterol

- Average diet contains about 300 mg of cholesterol per day.
- Body synthesizes about 700 mg/day.
- About 500mg of cholesterol is excreted through bile.
- Some of this partly reabsorbed from intestine.
- Remaining is converted to bile acids, which are excreted in the bile as bile

Degradation of cholesterol

- Cholesterol (50%) is converted to bile acids (excreted in feces), serves as a precursor for the synthesis of steroid hormones, vitamin D, coprostanol & cholestanol.
- The bile acids are synthesized in the liver from cholesterol.
- Contain 24 carbon atoms, 2 or 3 OH groups in steroid nucleus & a side chain ending in carboxyl group.

- Bile acids are amphipathic in nature.
- Possess both polar & non-polar groups.
- Serve as emulsifying agents in the intestine.
- Participate in digestion & absorption of lipids.
- Hydroxylation reaction:
- Cholesterol is converted to 7-hydroxy cholesterol by the action of the enzyme 7 α- hydroxylase (a microsomal enzyme).

- It is a rate limiting reaction.
- Incorporation of OH group at 7th carbon atom.
- A third OH group is added at carbon atom 12th in cholic acid.
- Chenodeoxycholic acid & cholic acid are primary bile acids.
- On conjugation with glycine or taurine, conjugated bile acids formed, namely Glycocholic acid & taurocholic acid

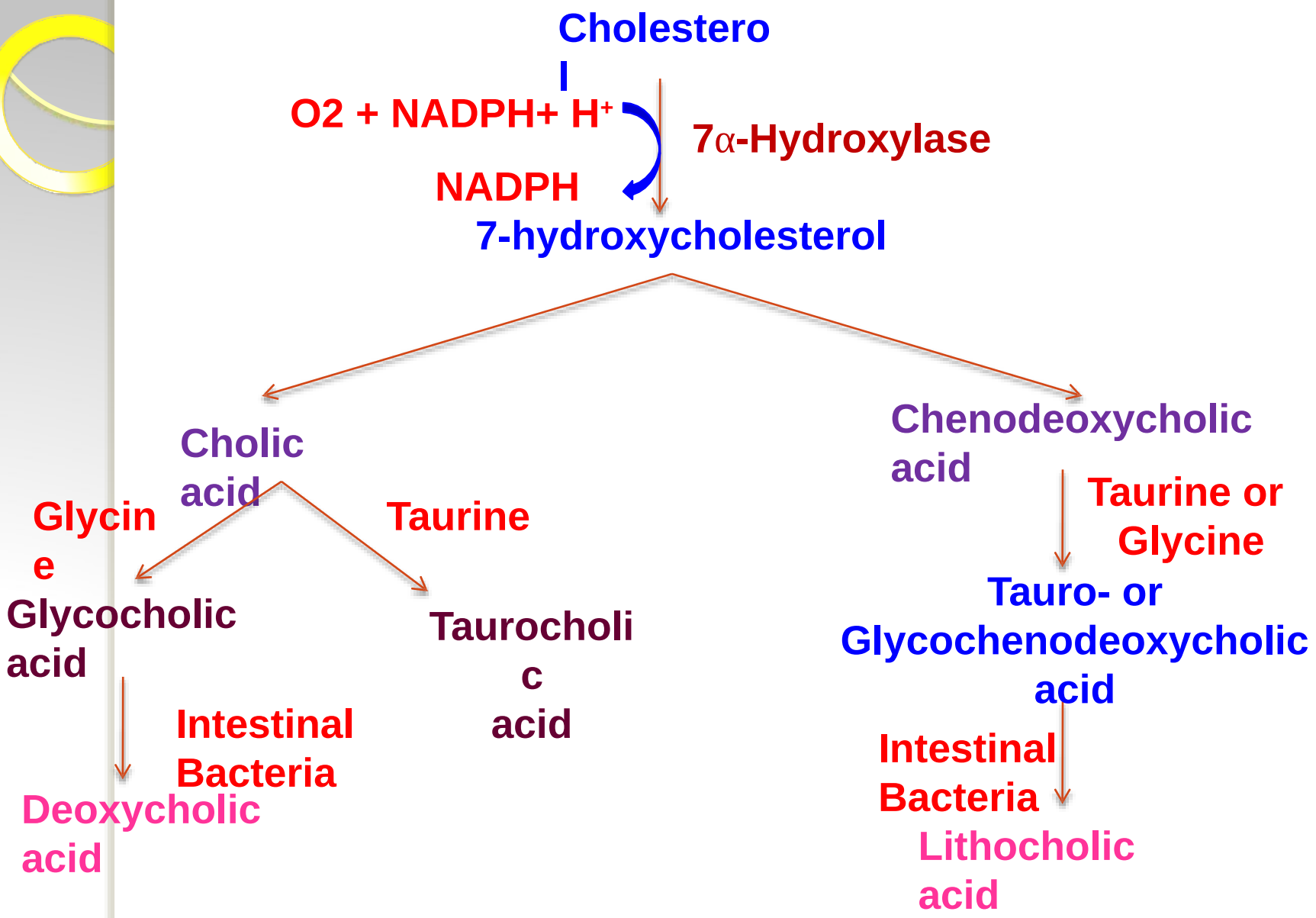
- These are more efficient in their function as surfactant.
- The conjugated bile acids are excreted through the bile.
- Bile salts:
- In the bile conjugated bile acids exist as sodium & potassium salts which are known as bile salts.

Secondary bile acids

- The primary bile acids are acted upon by the intestinal bacteria, which results in deconjugation & decarboxylation to form secondary bile acids.
- The deconjugated bile acids are partly converted to secondary bile acids by removal of OH group at 7th carbon.

- Cholic acid is converted to deoxycholic acid & chenodeoxycholic acid is converted to lithocholic acid.
- Deoxycholic acid & lithocholic acid are secondary bile acids.

Bile acid Synthesis



Enterohepatic circulation

- The conjugated bile salts synthesized in the liver accumulate in gall bladder.
- Secreted into the small intestine & they serve as emulsifying agents.
- A large portion of the bile salts (primary & secondary) are reabsorbed & returned to the liver through portal vein.

- The bile salts are recycled & reused several times in a day.
- This is known as enterohepatic circulation.
- About 15- 30 g of bile salts are secreted into the intestine each day & reabsorbed.
- Fecal excretion of bile salts is only route for removal of cholesterol from body.

Cholelithiasis

- Bile salts & phospholipids are responsible for keeping the cholesterol in bile in a soluble state.
- Due to bile salts & phospholipids deficiency (particularly bile salts), cholesterol crystals precipitate in the gall bladder often resulting in cholelithiasis- cholesterol gall stone disease.
- Cholelithiasis may be due to defective absorption of bile salts from the intestine, impairment in liver function, obstruction of biliary tract etc.

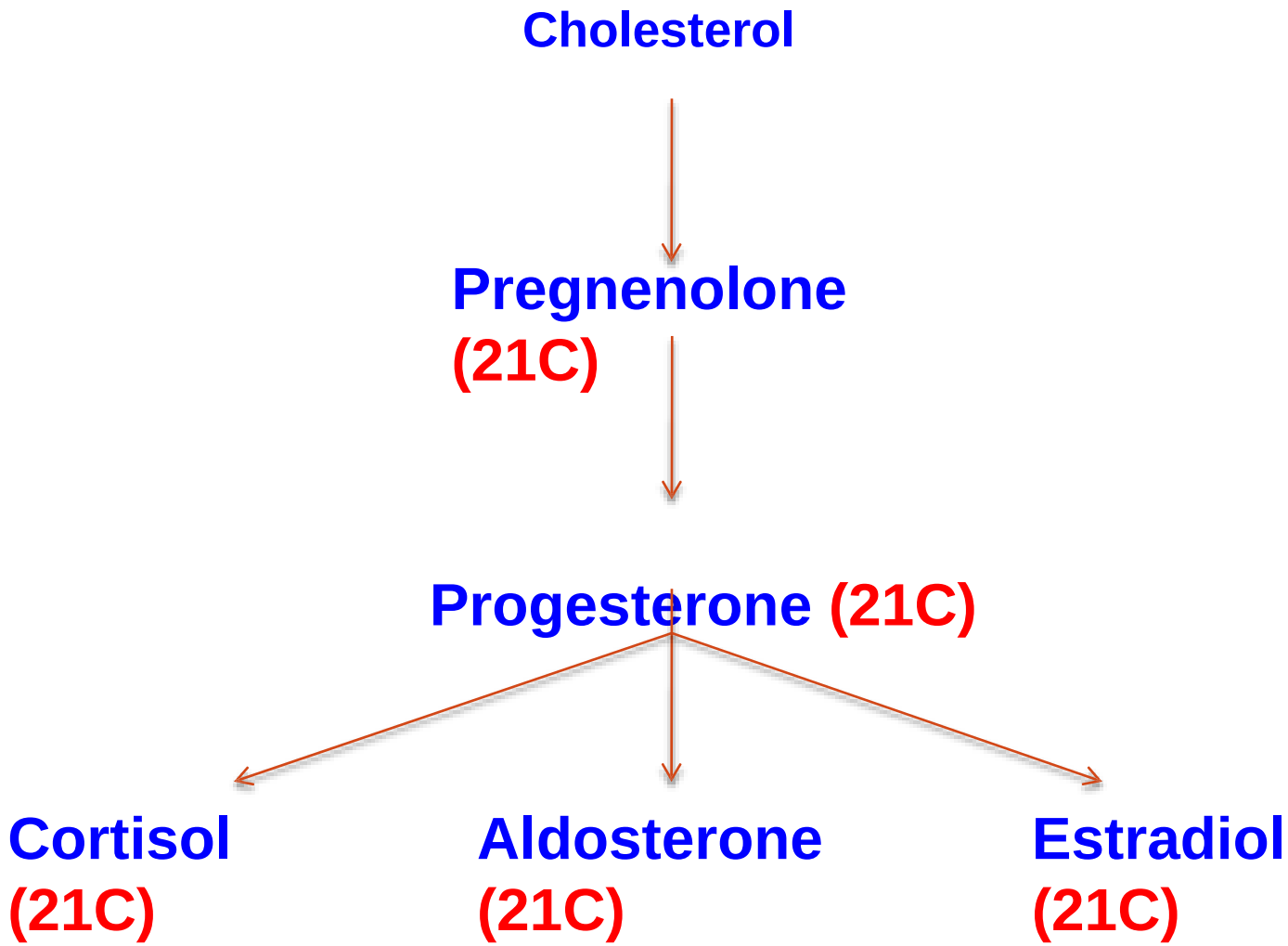
Synthesis of steroid hormones from cholesterol

- Cholesterol is the precursor for the synthesis of all the five classes of steroid hormones
- Glucocorticoids (Cortisol)
- Mineralocorticoides (Aldosterone)
- Progestins (Progesterone)
- Androgens (Testosterone)
- Estrogens (Estradiol)

Synthesis of vitamin D

- 7-Dehydrocholesteroal, an intermediate in the synthesis of cholesterol, is converted to cholecalciferol (vitamin D3) UV rays in the skin.

Synthesis of steroid hormones from cholesterol



Transport of cholesterol

- Cholesterol is present in the plasma lipoproteins in two forms
- About 70-75% of it is in an esterified form with long chain fatty acids.
- About 25-30% as free cholesterol.
- Free cholesterol readily exchanges between different lipoproteins & also with the cell membranes.

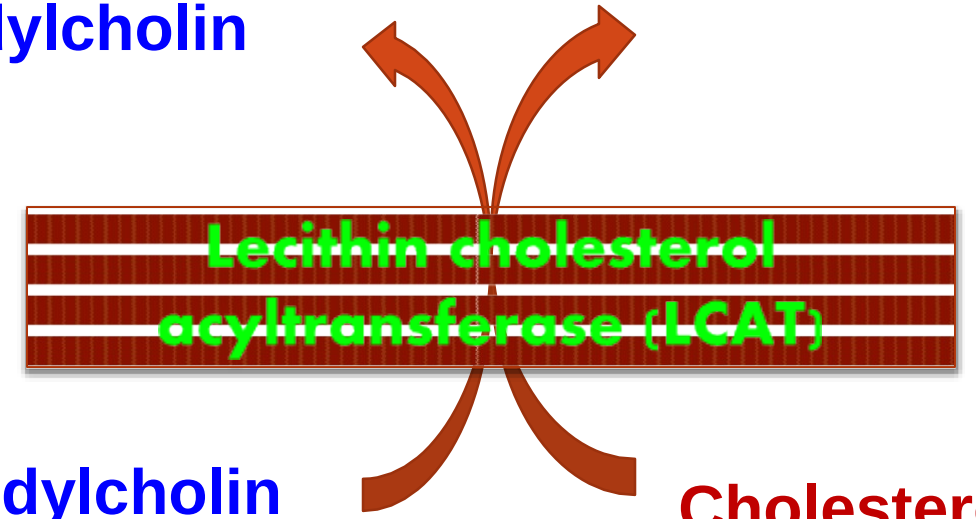
Role of LCAT

- High density lipoproteins (HDL) & the enzyme lecithin-cholesterol acyltransferase (LCAT) are responsible for the transport & elimination of cholesterol from the body.
- LCAT is a plasma enzyme, synthesized by the liver.

- It catalyses the transfer of fatty acid from the second position of phosphatidyl choline (lecithin) to the hydroxyl group of cholesterol.
- HDL-cholesterol is the real substrate for LCAT & this reaction is freely reversible.
- LCAT activity is associated with apo-A1 of HDL.

Reaction catalyzed by LCAT

Phosphatidylcholine



Cholesterol

Lysophosphatidylcholine

Cholesterol ester

Plasma cholesterol & its significance

- **Hypercholesterolemia:**
- **Increase in plasma cholesterol (>200mg/dl) is known as Hypercholesterolemia.**
- **It is observed in**
- **Diabetes mellitus:**
- **Due to increased cholesterol synthesis.**
- **The availability of acetyl CoA is increased.**

- **Hypothyroidism: Due to decrease in HDL receptors on hepatocytes.**
- **Obstructive jaundice:**
- **Due to an obstruction in the excretion of cholesterol through bile.**
- **Nephrotic syndrome:**
- **Due to increase in plasma lipoprotein fractions.**
- **Hypercholesterolemia is associated with atherosclerosis & coronary heart**

- Deposition of cholesterol esters & lipids in the intima of arterial walls leading to hardening of coronary arteries & cerebral blood vessels.
- **Bad cholesterol & good cholesterol:**
- LDLC is considered bad due to its involvement in atherosclerosis & related complications.

- LDLC may be regarded as lethally dangerous lipoprotein.
- HDLC cholesterol is good cholesterol.
- High concentrations counteracts atherogenesis.
- HDLC may be considered as highly desirable lipoprotein.
- HDLC-is good cholesterol
- LDLC-is bad cholesterol.

Control of hypercholesterolemia

- **Consumption of PUFA:** Dietary intake of PUFA reduces the plasma cholesterol levels.
- **Dietary cholesterol:** Cholesterol is found only in animal foods & not in plant foods.
- Dietary cholesterol influence on plasma cholesterol is minimal.
- Avoidance of cholesterol-rich foods is advocated to be on the safe side.

- **Plant sterols:** Certain plant sterols (sitostanol esters) & their esters reduce plasma cholesterol levels.
- They inhibit the intestinal absorption of dietary cholesterol.
- **Dietary fiber:** Fiber present in vegetables decreases the cholesterol absorption from the intestine.

- **Avoiding high carbohydrate diet:** Diets rich in carbohydrates (particularly sucrose) should be avoided to control hypercholesterolemia.
- **Impact of lifestyles:** Elevation in plasma cholesterol is observed in people with smoking, abdominal obesity, lack of exercise, stress, high blood pressure, consumption of soft water.

- **Lifestyle changes will reduce cholesterol levels.**
- **Moderate alcohol consumption:**
- **The beneficial effects of moderate alcohol intake are masked by the ill effects of chronic alcoholism.**
- **Red wine is beneficial due to its antioxidants, besides low alcohol content.**

Use of drugs

- Drugs such as lovastatin which inhibit HMC CoA reductase & decrease cholesterol synthesis.
- Statins currently in use include atorvastatin, simvastatin & pravastatin

References

- Lippincott's Illustrated Review: Biochemistry
- Textbook of Medical Biochemistry: Chatterjea
- Harper's Illustrated Biochemistry

Thank You